

Experiment – 03

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Subject: Advanced Database
Management System

Question 1:

Given a table containing information about bank deposits and withdrawals made using Paypal, write a query to retrieve the final account balance for each account, taking into account all the transactions recorded in the table with the assumption that there are no missing transactions.

transactions Table:

Column Name	Type
transaction_id	integer
account_id	integer
amount	decimal
transaction_type	varchar

transactions Example Input:

transaction_id	account_id	amount	transaction_type
123	101	10.00	Deposit
124	101	20.00	Deposit
125	101	5.00	Withdrawal
126	201	20.00	Deposit
128	201	10.00	Withdrawal

Example Output:

account_id	final_balance
101	25.00
201	10.00

Using account ID 101 as an example, \$30.00 was deposited into this account, while \$5.00 was withdrawn. Therefore, the final account balance can be calculated as the difference between the total deposits and withdrawals which is \$30.00 - \$5.00, resulting in a final balance of \$25.00.

The dataset you are querying against may have different input & output - this is just an example!

Solution:

```
SELECT
  ACCOUNT_ID,
  SUM(
    CASE
      WHEN TRANSACTION_TYPE = 'Deposit' THEN AMOUNT
      WHEN TRANSACTION_TYPE = 'Withdrawal' THEN -AMOUNT
    END
  ) AS FINAL_BALANCE
FROM TRANSACTIONS
GROUP BY ACCOUNT_ID;
```